Supplementary Tables

Supplementary Table S1. Arousal and bad epoch percentages in NREM/N3 sleep. Group-level percentages summarize epochs with arousals and visually annotated bad epochs across NREM and N3 sleep stages (mean ± SD). Data cover all NREM epochs and N3-only subsets by group, with N indicating contributing subjects per measure. The final column reports bad channels per group.

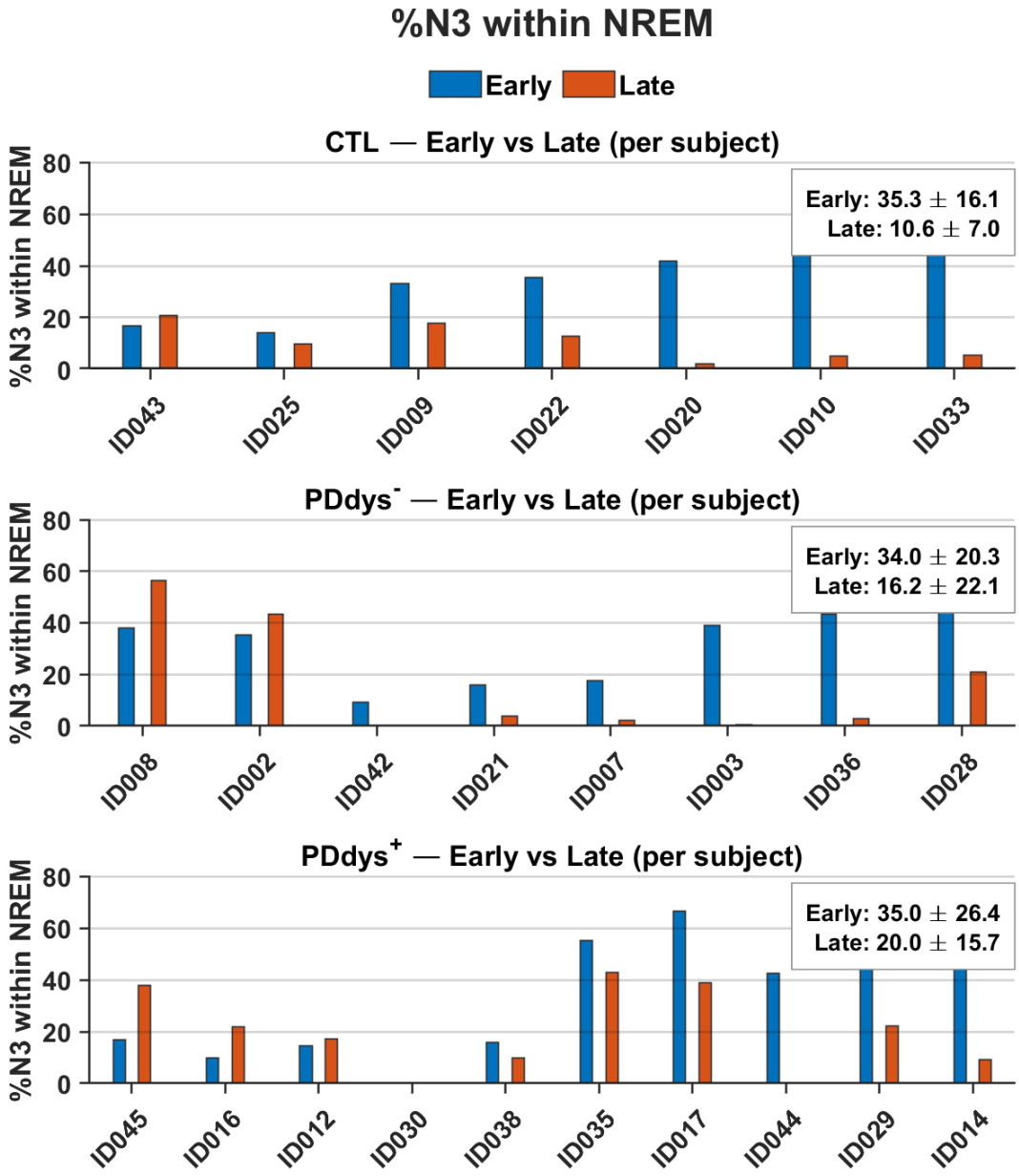
		Arousals		BadEpochs		BadChannels
Group	**N**	**NREM**	**N3**	**NREM**	**N3**	
CTL	8	12.1 ± 8.2	1.5 ± 1.8	4.9 ± 4.6	2.5 ± 5.2	3.5 ± 2.4
PD^{dys-}	9	5.4 ± 9.4	1.0 ± 2.6	8.0 ± 7.5	6.6 ± 7.0	5.9 ± 7.5
PD^{dys+}	11	9.7 ± 9.2	1.6 ± 2.4	15.2 ± 9.5	6.8 ± 8.5	6.0 ± 7.6

Supplementary Table S2. N3 epoch counts in Early and Late sleep phases and inclusion criteria. For each subject, the table reports the absolute number of artifact-free N3 epochs available in the Early (T2–T5) and Late (T6–T9) sleep phases, defined by NREM-based segmentation. The Included column indicates whether the subject met the minimum data requirement for the Early and Late analyses (i.e., at least 4 N3 epochs in each phase, equivalently $m=\min(n_{Early}, n_{Late}) \geq 4$) subjects not meeting this criterion were excluded from the corresponding analyses.

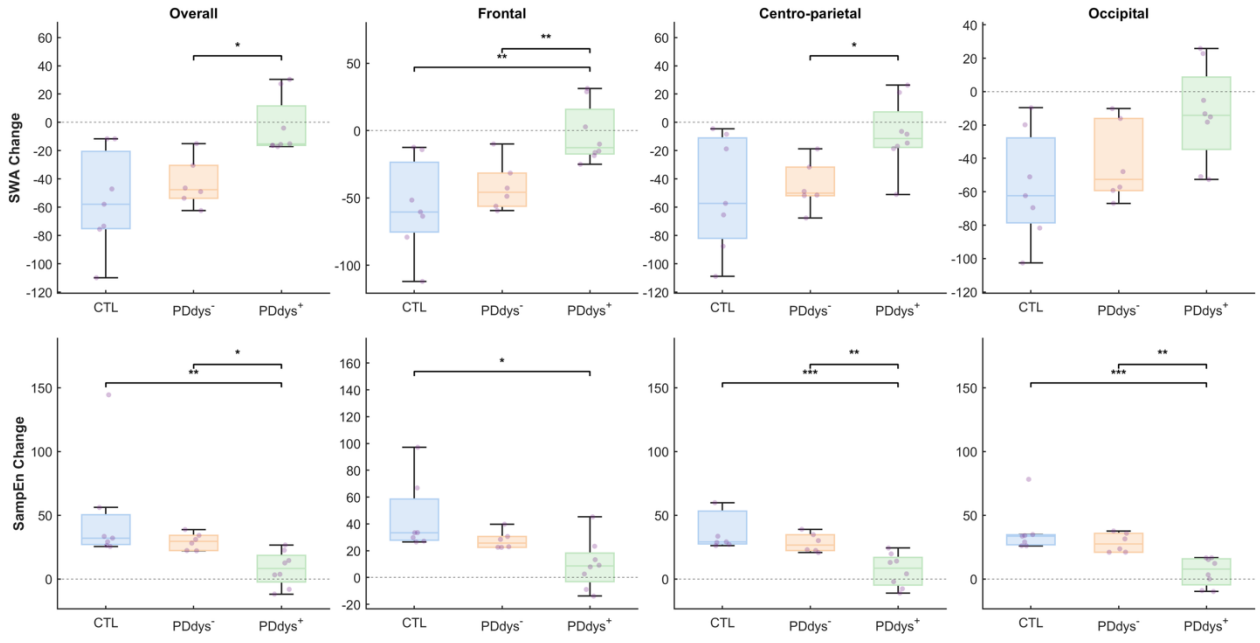
Group	Subj	nN3_E	nN3_L	Included
CTL	ID009	90	30	yes
CTL	ID010	75	43	yes
CTL	ID020	61	25	yes
CTL	ID022	42	15	yes
CTL	ID025	27	45	yes
CTL	ID033	110	9	yes
CTL	ID043	47	45	yes
PD^{dys-}	ID002	90	133	yes
PD^{dys-}	ID003	92	1	no
PD^{dys-}	ID007	32	4	yes
PD^{dys-}	ID008	67	79	yes
PD^{dys-}	ID021	24	19	yes
PD^{dys-}	ID028	142	81	yes
PD^{dys-}	ID036	117	5	yes
PD^{dys-}	ID042	13	0	no
PD^{dys+}	ID014	150	19	yes
PD^{dys+}	ID016	10	22	yes
PD^{dys+}	ID017	200	92	yes
PD^{dys+}	ID029	204	50	yes
PD^{dys+}	ID030	0	0	no
PD^{dys+}	ID035	76	52	yes
PD^{dys+}	ID038	53	14	yes
PD^{dys+}	ID044	63	0	no
PD^{dys+}	ID045	17	38	yes

Supplementary Figures

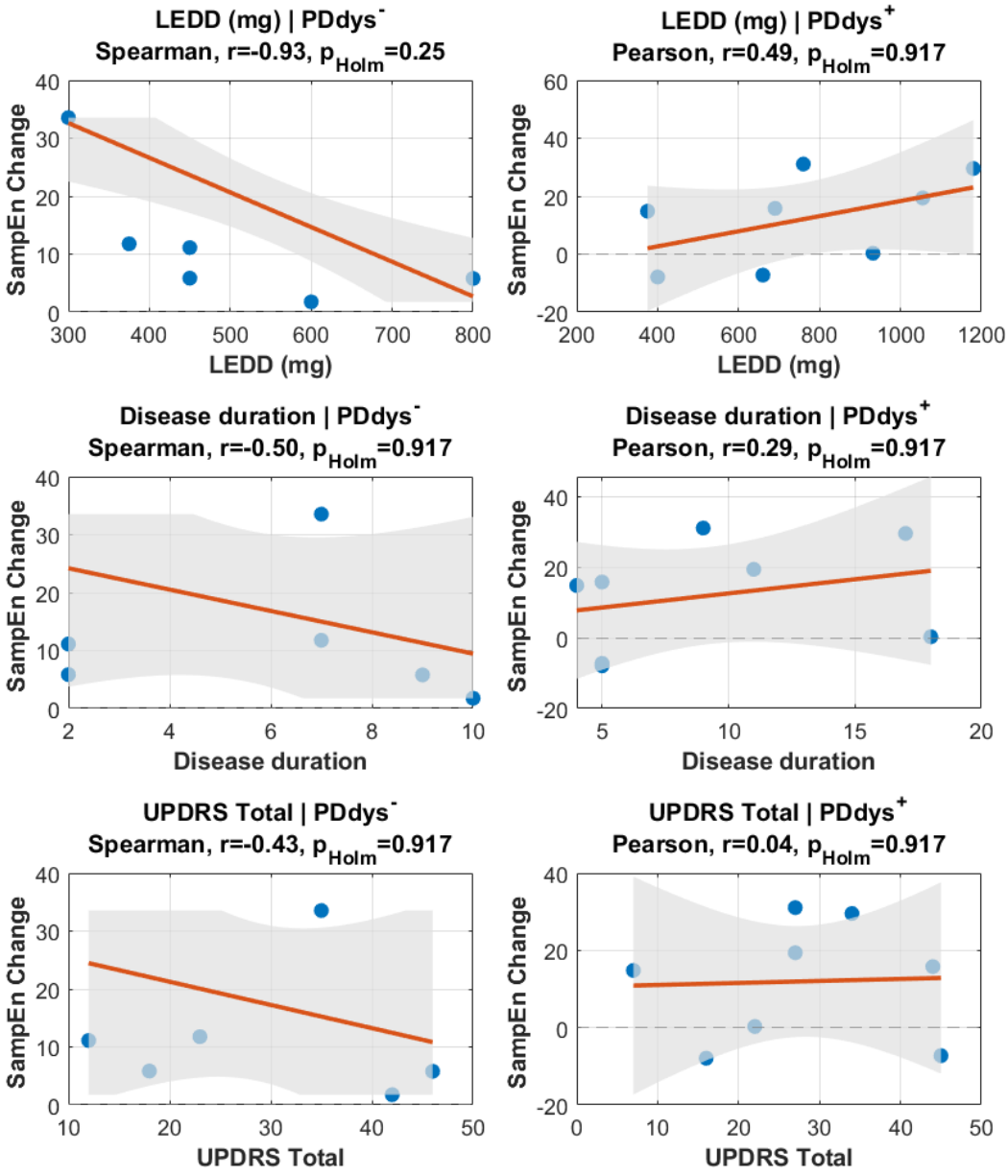
Supplementary Figure S1. N3 epoch percentages within NREM sleep Early/Late.
For each group, bar plots show the percentage of N3 epochs within NREM sleep for each subject, computed separately for the Early (blue bars) and Late (orange bars) sleep phases defined by NREM-based segmentation. The box in the upper-right of each panel reports the group mean ± SD of %N3 within NREM for Early and Late, summarized across subjects.



Supplementary Figure S2. Late–Early SWA and SampEn changes by region of interest. Box plots show the normalized overnight changes $(Late - Early) / Early$ in SWA and SampEn aggregated at region of interest (ROI) levels (i.e., Frontal, Centro-parietal and Occipital) across groups (N=21). Normality was assessed with the Shapiro–Wilk test; frontal and occipital SWA were analyzed with one-way ANOVA, whereas all other comparisons used Kruskal–Wallis tests, followed by post-hoc pairwise contrasts with effect size estimation, Tukey after ANOVA, Dunn after Kruskal wallis. Significant group effects were observed for all panels (SWA: p = 0.003–0.029; SampEn: p < 0.05–0.001), with effect sizes ranging from small to large (0.24–0.71). Post-hoc tests indicated a reduced overnight downscaling of SWA in PD^{dys+} compared with CTL and PD^{dys-}, with several contrasts not surviving correction or yielding small-to-moderate effects (e.g., overall SWA CTL versus PD^{dys-} p = 0.53, CTL versus PD^{dys+} p = 0.08). In contrast, SampEn changes robustly differentiated PD^{dys+} from both CTL and PD^{dys-} across most ROIs, except for the frontal region; asterisks denote significance levels (* p < 0.05, ** p < 0.01, ** p < 0.001).



Supplementary Figure S3. Correlations between frontal N3 SampEn dynamics and disease-related clinical measures in PD groups. Scatterplots show associations between frontal N3 normalized Late–Early SampEn changes and levodopa equivalent daily dose (LEDD), disease duration, and MDS-UPDRS Total score in PD^{dys-} and PD^{dys+} groups. Each point represents one subject; solid lines indicate fitted regression lines with shaded 95% prediction intervals. Correlation coefficients reflect Pearson's r (normality confirmed via Shapiro–Wilk) or Spearman's ρ otherwise, with p-values adjusted via Holm correction. No associations reached statistical significance (all $p > 0.05$).



Supplementary Figure S4. Correlations between frontal N3 SampEn dynamics and sleep-related clinical measures across groups. Scatterplots show associations between frontal N3 normalized Late–Early SampEn changes and Pittsburgh Sleep Quality Index (PSQI), Epworth Sleepiness Scale (ESS), and sleep efficiency in CTL, PD^{dys-}, and PD^{dys+} groups. Each point represents one subject; solid lines indicate fitted regression lines with shaded 95% prediction intervals. Correlation coefficients reflect Pearson's r (normality confirmed via Shapiro–Wilk) or Spearman's ρ otherwise, with p-values adjusted via Holm correction. No associations reached statistical significance (all $p > 0.05$).

